

CORE 2

Motion in 2D - Projectile Motion

Source-question custody: selected demand anchors plus the retained non-selected Q15 candidate.

Custody boundary

These are faithful non-verbatim source-derived records. They preserve identity, conditions, answer custody, and the primary capability. They are not authored hints and are not automatically Grade 9 learner-eligible.

Q28 - Selected demand

JEE Main 2026 | 04 Apr | Shift 2 | Q29

Governed faithful restatement

A projectile has $x=24t$ and $y=43.6t-4.9t^2$ (SI units). Find the angle its velocity vector makes with the horizontal at $t=2$ s.

Conditions

No additional conditions in the governed record.

Source-answer custody

Answer

45 degrees.

Reasoning custody

1. Differentiate: $v_x=24$ and $v_y=43.6-9.8t$.
2. At $t=2$ s, $v_y=24$ m/s.
3. $\tan(\theta)=v_y/v_x=1$, so $\theta=45$ degrees.

Primary capability preserved: CAP-KIN-2D-INDEPENDENT-COMPONENTS

Q15 - Selected demand

JEE Advanced 2018 | Paper 2 | Q8

Governed faithful restatement

A ball is launched from level ground at 45 degrees and reaches maximum height 120 m. At the first ground impact it loses half its kinetic energy and rebounds at 30 degrees above horizontal. Find the new maximum height.

Conditions

Same-level first flight; half kinetic energy lost at first bounce; post-bounce launch angle 30 degrees.

Source-answer custody

Answer

30 m.

Reasoning custody

1. Initial maximum height gives $u_y^2 = 2g(120)$. At 45 degrees, $u^2 = 480g$.
2. After losing half the kinetic energy, $v^2 = u^2/2 = 240g$.
3. New $v_y^2 = v^2 \sin^2(30 \text{ degrees}) = 60g$, hence $H = 60g/(2g) = 30 \text{ m}$.

Primary capability preserved: CAP-KIN-PROJECTILE-MODEL

Q21 - Selected demand

JEE Advanced 2022 | Paper 1 | Q8

Governed faithful restatement

A projectile has ordinary range d . At the highest point it enters a region where effective downward acceleration becomes $g/0.81$. If the new total range is $d'=n d$, find n .

Conditions

Acceleration changes only after the apex; horizontal velocity is unchanged at the boundary.

Source-answer custody

Answer

0.95.

Reasoning custody

1. The apex is at $d/2$.
2. The apex height is unchanged; with $g/0.81$ after the apex, the descent time is 0.9 of the ordinary descent time.
3. The second horizontal segment is $0.9(d/2)$, so $d'=d/2+0.9d/2=0.95d$.

Primary capability preserved: CAP-KIN-PROJECTILE-MODEL

Q26 - Selected demand

JEE Advanced 2026 | Paper 1 | Q6

Governed faithful restatement

A projectile from O passes through $P=(5\text{ m}, 1\text{ m})$. Source statements ask about the required speed and whether maximum height occurs before or after P for specified launch angles.

Conditions

Uniform gravity downward; $P=(5,1)$ relative to O.

Source-answer custody

Answer

Source-custody answer: options (A) and (B).

Reasoning custody

1. Use $y=x \tan(\theta) - g x^2 / [2v^2 \cos^2(\theta)]$ at P.
2. For $\theta=45$ degrees, $v^2=25g/4$, so $v=5\sqrt{g}/2$.
3. The local trajectory slope at $x=5$ is negative, so the apex is before P.

Primary capability preserved: CAP-KIN-PROJECTILE-MODEL

Q23 - Selected demand

JEE Advanced 2025 | Paper 2 | Q15

Governed faithful restatement

A 0.2 kg projectile is launched at 270 m/s at 60 degrees in a medium with linear drag $F = -c v$, $c = 0.1$ kg/s. It hits a vertical wall after 2 s. Using $e = 2.7$ as specified by the source, find the horizontal distance.

Conditions

Linear viscous drag opposes instantaneous velocity; source uses $e = 2.7$.

Source-answer custody

Answer

170 m (official accepted interval 167-171 m).

Reasoning custody

1. Horizontal motion obeys $m \frac{dv_x}{dt} = -c v_x$.
2. $v_{x0} = 270 \cos 60 = 135$ m/s and $m/c = 2$ s.
3. $x = v_{x0}(m/c)[1 - \exp(-ct/m)] = 270(1 - 1/e) = 170$ m at $t = 2$ s.

Primary capability preserved: CAP-KIN-2D-CONSTANT-ACCELERATION

Q15 - Retained non-selected comparison evidence

JEE Advanced 2023 | Paper 1 | Q1

Governed faithful restatement

A ball slides frictionlessly from height h above a terrace, leaves horizontally, strikes the ground, and rebounds with coefficient of restitution $1/\sqrt{3}$. Multiple source statements concern launch speed, impact angle, rebound velocity, and rebound geometry.

Conditions

This is a building/slide/restitution multi-statement problem, not the owner Q15 energy-loss/30-degree rebound demand.

Source-answer custody

Answer

Source-custody answer: options (A), (C), and (D).

Reasoning custody

1. Energy on the slide gives $u_0 = \sqrt{2gh}$.
2. Impact geometry gives 60-degree impact angle.
3. The restitution model determines the rebound vertical component and rebound-height/range relation.

Primary capability preserved: CAP-KIN-PROJECTILE-MODEL